

The Playlist Operator

a matrix that plays music, whose picture is an image of your choice

1 What we want to be true

Take N songs, each a vector of audio samples $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^D$ (in the current build $N = 6$ and $D \approx 5.5 \times 10^5$ at 1 kHz). We want one matrix $\mathbf{A} \in \mathbb{R}^{D \times D}$ that *is* the playlist:

$$\mathbf{A} \mathbf{x}_n = \mathbf{x}_{n+1} \quad (n = 1, \dots, N-1), \quad \mathbf{A} \mathbf{x}_N = \mathbf{x}_1. \quad (1)$$

Seed it with any song and repeated multiplication plays the playlist, forever, in a cycle.

Collect the songs as the columns of $\mathbf{X} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_N] \in \mathbb{R}^{D \times N}$, and let $\mathbf{S} \in \mathbb{R}^{N \times N}$ be the *cyclic shift* permutation, e.g. for $N = 4$:

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{X} \mathbf{S} = [\mathbf{x}_2 \ \mathbf{x}_3 \ \mathbf{x}_4 \ \mathbf{x}_1]. \quad (2)$$

(Column n of $\mathbf{X} \mathbf{S}$ is the successor of song n ; the top-right 1 closes the loop.) All N statements in (1) then compress into one matrix equation:

$$\boxed{\mathbf{A} \mathbf{X} = \mathbf{X} \mathbf{S}} \quad (3)$$

The key count. Equation (3) imposes DN scalar constraints on the D^2 unknown entries of $\mathbf{A}$, and $N \ll D$. The songs span only an N -dimensional subspace $\mathcal{S} = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_N) \subset \mathbb{R}^D$, so (3) dictates how $\mathbf{A}$ acts on $\mathcal{S}$ and says *nothing* about the orthogonal complement $\mathcal{S}^\perp$, which has dimension $D - N$. Roughly $D(D - N) \approx 3 \times 10^{11}$ degrees of freedom remain. That freedom is where the artwork will live.

2 Solving $\mathbf{A} \mathbf{X} = \mathbf{X} \mathbf{S}$

2.1 Warm-up: projecting onto song space, and why $\mathbf{G} = \mathbf{X}^\top \mathbf{X}$ appears

Given an arbitrary vector $\mathbf{v} \in \mathbb{R}^D$, which combination of songs $\mathbf{X} \mathbf{c}$ (coefficients $\mathbf{c} \in \mathbb{R}^N$) is closest to it? Minimize the squared distance, expanding it fully:

$$\begin{aligned} J(\mathbf{c}) &= \|\mathbf{X} \mathbf{c} - \mathbf{v}\|^2 = (\mathbf{X} \mathbf{c} - \mathbf{v})^\top (\mathbf{X} \mathbf{c} - \mathbf{v}) \\ &= \mathbf{c}^\top \mathbf{X}^\top \mathbf{X} \mathbf{c} - \mathbf{c}^\top \mathbf{X}^\top \mathbf{v} - \mathbf{v}^\top \mathbf{X} \mathbf{c} + \mathbf{v}^\top \mathbf{v} \\ &= \mathbf{c}^\top \mathbf{G} \mathbf{c} - 2 \mathbf{c}^\top \mathbf{X}^\top \mathbf{v} + \|\mathbf{v}\|^2, \quad \mathbf{G} \equiv \mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{N \times N}, \end{aligned} \quad (4)$$

using that $\mathbf{v}^\top \mathbf{X} \mathbf{c}$ is a scalar, hence equal to its own transpose $\mathbf{c}^\top \mathbf{X}^\top \mathbf{v}$. Setting the gradient to zero,

$$\nabla_{\mathbf{c}} J = 2 \mathbf{G} \mathbf{c} - 2 \mathbf{X}^\top \mathbf{v} = \mathbf{0} \implies \mathbf{G} \mathbf{c} = \mathbf{X}^\top \mathbf{v} \implies \mathbf{c}^* = \mathbf{G}^{-1} \mathbf{X}^\top \mathbf{v}. \quad (5)$$

$\mathbf{G}$ is the *Gram matrix* of the playlist: $G_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$ holds every pairwise song correlation. It is invertible iff the songs are linearly independent (for real recordings, always; measured condition number here: 1.01 — long audio vectors are nearly orthogonal). The matrix $\mathbf{X}^\top \mathbf{v}$ *correlates* the input with each song; $\mathbf{G}^{-1}$ *unmixes* the overlap between songs.

Reassembling the best approximation $\mathbf{X}\mathbf{c}^*$ defines the orthogonal projector onto song space and its complement:

$$\mathbf{P} = \mathbf{X}\mathbf{G}^{-1}\mathbf{X}^\top, \quad \mathbf{P}_\perp = \mathbf{I} - \mathbf{X}\mathbf{G}^{-1}\mathbf{X}^\top, \quad \mathbf{P}_\perp \mathbf{x}_n = \mathbf{0} \text{ for every song.} \quad (6)$$

$$(\mathbf{P}_\perp \mathbf{X} = \mathbf{X} - \mathbf{X}\mathbf{G}^{-1}\mathbf{G} = \mathbf{X} - \mathbf{X} = \mathbf{0}.)$$

2.2 The minimum-norm solution, by setting a gradient to zero

Among all solutions of (3) we single out the smallest one — minimum Frobenius norm ($\|\mathbf{A}\|_F^2 = \sum_{ij} A_{ij}^2$):

$$\mathbf{A}_0 = \arg \min_{\mathbf{A}} \frac{1}{2} \|\mathbf{A}\|_F^2 \quad \text{subject to} \quad \mathbf{A}\mathbf{X} = \mathbf{X}\mathbf{S}. \quad (7)$$

Introduce a Lagrange multiplier matrix $\mathbf{\Lambda} \in \mathbb{R}^{D \times N}$ (one multiplier per scalar constraint) and form the Lagrangian

$$\mathcal{L}(\mathbf{A}, \mathbf{\Lambda}) = \frac{1}{2} \|\mathbf{A}\|_F^2 - \text{tr}(\mathbf{\Lambda}^\top (\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{S})). \quad (8)$$

Set the gradient with respect to $\mathbf{A}$ to zero (using $\nabla_{\mathbf{A}} \frac{1}{2} \|\mathbf{A}\|_F^2 = \mathbf{A}$ and $\nabla_{\mathbf{A}} \text{tr}(\mathbf{\Lambda}^\top \mathbf{A}\mathbf{X}) = \mathbf{\Lambda}\mathbf{X}^\top$):

$$\nabla_{\mathbf{A}} \mathcal{L} = \mathbf{A} - \mathbf{\Lambda}\mathbf{X}^\top = \mathbf{0} \implies \mathbf{A} = \mathbf{\Lambda}\mathbf{X}^\top. \quad (9)$$

So the optimum necessarily has rank $\leq N$ and row space inside $\text{span}(\mathbf{X})$ — the constraint only “pays” for structure aligned with the songs; any component orthogonal to them would add norm without helping. Substitute (9) into the constraint to find $\mathbf{\Lambda}$:

$$\mathbf{\Lambda} \underbrace{\mathbf{X}^\top \mathbf{X}}_{\mathbf{G}} = \mathbf{X}\mathbf{S} \implies \mathbf{\Lambda} = \mathbf{X}\mathbf{S}\mathbf{G}^{-1} \implies \boxed{\mathbf{A}_0 = \mathbf{X}\mathbf{S}\mathbf{G}^{-1}\mathbf{X}^\top} \quad (10)$$

Check it, step by step:

$$\mathbf{A}_0 \mathbf{X} = \mathbf{X}\mathbf{S}\mathbf{G}^{-1} \underbrace{\mathbf{X}^\top \mathbf{X}}_{=\mathbf{G}} = \mathbf{X}\mathbf{S} \underbrace{\mathbf{G}^{-1}\mathbf{G}}_{=\mathbf{I}_N} = \mathbf{X}\mathbf{S}. \quad \checkmark \quad (11)$$

Read $\mathbf{A}_0 \mathbf{v} = \mathbf{X}\mathbf{S}(\mathbf{G}^{-1}\mathbf{X}^\top \mathbf{v})$ right to left: correlate the input with every song ($\mathbf{X}^\top \mathbf{v}$), unmix ($\mathbf{G}^{-1}$, exactly the least-squares coefficients (5)), advance one playlist step ($\mathbf{S}$), rebuild audio ($\mathbf{X}$).

2.3 The full solution family

If $\mathbf{A}$ and $\mathbf{A}'$ both solve (3), their difference kills every song: $(\mathbf{A} - \mathbf{A}')\mathbf{X} = \mathbf{0}$. A matrix annihilates $\mathbf{S}$ exactly when it factors through the projector $\mathbf{P}_\perp$. Hence the complete solution set is the affine family

$$\boxed{\mathbf{A} = \mathbf{A}_0 + \mathbf{Z}\mathbf{P}_\perp, \quad \mathbf{Z} \in \mathbb{R}^{D \times D} \text{ arbitrary.}} \quad (12)$$

The free part is *inaudible by construction*: $\mathbf{Z}\mathbf{P}_\perp \mathbf{x}_n = \mathbf{Z}\mathbf{0} = \mathbf{0}$, so playback is exact for every choice of $\mathbf{Z}$ — the code measures $\max_n \|\mathbf{A}\mathbf{x}_n - \mathbf{x}_{n+1}\| / \|\mathbf{x}_{n+1}\| \approx 10^{-14}$, i.e. floating-point roundoff, always.

3 From matrix to picture: blocks, pooling, and two honest views

$\mathbf{A}$ has $D^2 \approx 3 \times 10^{11}$ entries; a print has $P \times P \approx 10^6$ pixels. Each pixel must summarize a $k \times k$ block of adjacent matrix entries, $k = \lfloor D/P \rfloor$ — block (p, q) means rows $pk, \dots, (p+1)k - 1$ and columns $qk, \dots, (q+1)k - 1$ of $\mathbf{A}$, i.e. “what the operator does to time-window q of the input, as heard in time-window p of the output.”

Averaging over blocks is one matrix operation. Define the *pooling matrix* $\mathbf{\Pi} \in \mathbb{R}^{P \times D}$: row p contains $1/k$ over block p ’s samples and 0 elsewhere. A toy example with $D = 6$, $P = 2$, $k = 3$:

$$\mathbf{\Pi} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}, \quad (13)$$

so $\mathbf{\Pi v}$ averages consecutive triples of $\mathbf{v}$. $\mathbf{\Pi}$ is fully determined by two numbers: the signal length D and the chosen print resolution P . Pooling rows *and* columns of $\mathbf{A}$ gives the $P \times P$ **block-mean picture**

$$\mathbf{M} = \mathbf{\Pi A \Pi}^\top, \quad M_{pq} = \frac{1}{k^2} \sum_{i \in \text{block } p} \sum_{j \in \text{block } q} A_{ij}. \quad (14)$$

Every pixel is *exactly* the average of its block of true matrix entries — the canonical downsampling of an image too large to print, not an approximation with error. A second, equally exact per-block statistic is the **block energy** (root-mean-square), $E_{pq} = \frac{1}{k^2} \sum_{i \in p, j \in q} A_{ij}^2$.

Why ever look at two views of one matrix? Audio oscillates around zero, so within any block the entries of $\mathbf{A}_0$ nearly cancel: the mean picture of $\mathbf{A}_0$ is almost blank, while its squares do not cancel — the energy picture is where the music’s own block structure (loudness correlations, silence bands) is visible. Both are honest; they answer different questions. The printed artwork uses the mean channel (14), because it is *linear* in $\mathbf{A}$ — and that linearity is what the next section exploits. Neither is ever computed from the dense $\mathbf{A}$: with $\mathbf{A} = \mathbf{L R}^\top$ kept factored, $\mathbf{M} = (\mathbf{\Pi L})(\mathbf{\Pi R})^\top$ and E_{pq} streams through small per-block Gram matrices.

4 Choosing the picture

Substitute the family (12), with a low-rank free part $\mathbf{Z} = \mathbf{U V}^\top$ ($\mathbf{U}, \mathbf{V} \in \mathbb{R}^{D \times q}$), into the mean picture (14); linearity separates music from freedom:

$$\mathbf{M} = \underbrace{\mathbf{\Pi A}_0 \mathbf{\Pi}^\top}_{\mathbf{M}_0 \text{ (fixed by the music, nearly blank)}} + (\mathbf{\Pi U})(\mathbf{\Pi P}_\perp \mathbf{V})^\top. \quad (15)$$

A target image $\mathbf{T}$ has $P^2 \approx 10^6$ numbers; $\mathbf{Z}$ has $\approx 3 \times 10^{11}$ inaudible degrees of freedom. The counting is overwhelming: (almost) *any* picture is reachable. We want

$$(\mathbf{\Pi U})(\mathbf{\Pi P}_\perp \mathbf{V})^\top = \mathbf{T}_{\text{amp}} - \mathbf{M}_0 \equiv \mathbf{\Delta}, \quad (16)$$

where $\mathbf{T}_{\text{amp}}$ is the target after centering and amplitude scaling (a display is affine-invariant, so nothing is lost). Three steps:

1. Compress the image (SVD). The truncated singular value decomposition

$$\mathbf{\Delta} \approx \mathbf{W}_q \mathbf{\Sigma}_q \mathbf{Y}_q^\top = (\mathbf{W}_q \mathbf{\Sigma}_q^{1/2})(\mathbf{Y}_q \mathbf{\Sigma}_q^{1/2})^\top \equiv \tilde{\mathbf{U}} \tilde{\mathbf{V}}^\top, \quad \tilde{\mathbf{U}}, \tilde{\mathbf{V}} \in \mathbb{R}^{P \times q}, \quad (17)$$

is, by the Eckart–Young theorem, the best rank- q approximation of the residual image. Its role is purely *compression* — writing the picture as q outer products so that $\mathbf{Z}$ stays storable ($D \times q$ instead of $D \times D$). Smooth cloud images compress to small q (~ 150 , error $< 1\%$); images full of sharp lines need more.

2. Lift pixels to samples. We need sample-level columns with prescribed pooled values: $\Pi U = \tilde{U}$. Any *right inverse* $E \in \mathbb{R}^{D \times P}$ of the pooling matrix ($\Pi E = I_P$) does this. The simplest is “hold each pixel value constant across its k samples” ($E = k \Pi^\top$; averaging a constant returns it). We use a smooth variant — linear interpolation between block centers plus a piecewise-constant correction — which still satisfies $\Pi E = I_P$ exactly but avoids blocky columns. Set $U = E\tilde{U}$, $V = E\tilde{V}$.

3. Absorb the projector (a fixed-point iteration). The picture uses $\Pi P_\perp V$, not ΠV :

$$\Pi P_\perp V = \Pi V - \Pi X G^{-1} X^\top V = \tilde{V} - \underbrace{\Pi X G^{-1} X^\top V}_{\text{rank} \leq N, \text{ small}}. \quad (18)$$

The disturbance is the pooled shadow of a rank- N matrix, and it is small because generic image columns are nearly orthogonal to the N specific audio signals. Remove it by iterating

$$V \leftarrow V + E(\tilde{V} - \Pi P_\perp V). \quad (19)$$

Writing the pooled error $e = \tilde{V} - \Pi P_\perp V$, one step of (19) maps $e \leftarrow \Pi X G^{-1} X^\top E e$: the error is fed through an operator of rank $\leq N$ and norm well below one, so it shrinks geometrically; two or three iterations reach sub-percent accuracy. (This is *iterative refinement* — driving a residual to zero, as in Richardson iteration for linear systems — not power iteration, which does the opposite: amplifies a dominant eigenvector.)

Because V enters the operator only through $P_\perp V$, playback remains exact at every iterate; measured: image error $< 1\%$, playback error $\sim 10^{-14}$, simultaneously.

5 The pixel-exact song print: an error budget, and why looping is a stability question

The second construction drops pooling entirely. One song is cut into N consecutive windows of n samples, the columns of $W \in \mathbb{R}^{n \times N}$ (unit-normalised), and the same derivation as before gives the cyclic *window-advance* operator

$$A_0 = W S G^{-1} W^\top \in \mathbb{R}^{n \times n}, \quad A_0 w_k = w_{k+1}, \quad A_0 w_N = w_1. \quad (20)$$

A_0 is $n \times n$ however many windows there are, so with n equal to the print side it is displayed *one matrix entry per pixel* — no pooling, nothing to be honest *about*. Every design variable is tied together by one count of samples: a song of duration T at sample rate f has $fT = Nn = \rho n^2$ samples, where $\rho = N/n$ is the rank fraction — the fraction of the n^2 pixels carrying independent audio. Hence $n = \sqrt{fT/\rho}$, and at $\rho = 1$ each pixel is worth exactly one sample. Real audio never quite reaches $\rho = 1$: adjacent windows correlate, G degenerates, and (measured) playback survives to $\rho \approx 0.95$. Digital silence is a separate, harder wall — no linear map sends $\mathbf{0}$ to the music that follows it — cured by an inaudible dither floor.

Pricing an error budget. Suppose we accept playback error up to a tolerance ε . What freedom does that buy? Take the SVD $A_0 = \sum_{i=1}^N \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ and attenuate component i by $a_i \in [0, 1]$ (so $a_i = 1$ deletes it). The perturbed operator sends window k to

$$A' w_k = w_{k+1} - \sum_i a_i \sigma_i \mathbf{u}_i (\mathbf{v}_i^\top w_k), \quad \text{so} \quad \|A' w_k - w_{k+1}\|^2 = \sum_i a_i^2 \sigma_i^2 (\mathbf{v}_i^\top w_k)^2 \quad (21)$$

by orthonormality of the $\mathbf{u}_i$. The cost of any attenuation pattern is a closed formula — no reconstruction needed — and it is exactly linear in a global scaling of $\mathbf{a}$, so projecting a candidate into the budget is a single division. Measured on a real song: $\varepsilon = 10^{-4}$ buys *no* whole component (the cheapest costs 3×10^{-4}); $\varepsilon = 10^{-1}$ buys the 366 smallest of 950 — which change the picture imperceptibly, because ink is dominated by the large- σ components.

Why looping is not an error bound. The static price above controls *one* application. Playing the song on repeat iterates $\mathbf{A}$ thousands of times, and there the exact operator has a hidden virtue: its column space is the window space (every column of $\mathbf{A}_0$ is a combination of shifted windows), and *on* that space it acts as the pure cyclic shift $\mathbf{S}$ in coefficients. Its eigenvalues are the N -th roots of unity together with 0 — all on or inside the unit circle — so nothing grows, ever: the song loops forever. An attenuated $\mathbf{A}'$ loses this: its eigenvalues drift off the circle, and any eigenvalue $1 + \delta$ multiplies drift by $(1 + \delta)^t$ over t steps, amplified further through the operator’s gain ($\sigma_{\max} \sim 3 \times 10^3$). Measured: a truncation with static error 10^{-2} per window *diverges to overflow within one pass* through the song. “It should faithfully loop” is therefore a *spectral* constraint — eigenvalues must stay on the unit circle — not a tolerance on single-step error; the optimizer checks a full pass numerically for every candidate.

Freedom that provably loops. There is a perturbation family that costs *nothing*, statically or dynamically. Split $\mathbb{R}^n$ into window space and its orthogonal complement, and add

$$\mathbf{A} = \mathbf{A}_0 + \mathbf{Y} \mathbf{P}_\perp, \quad \text{range}(\mathbf{Y}) \subseteq \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_N), \quad (22)$$

with $\mathbf{P}_\perp$ the projector onto the complement. Then $\mathbf{A}\mathbf{w}_k = \mathbf{w}_{k+1}$ exactly ($\mathbf{P}_\perp$ kills every window), and in the split basis $\mathbf{A}$ is block upper triangular with the exact shift in one diagonal block and $\mathbf{0}$ in the other — its eigenvalues are *still* the roots of unity and 0. Exact playback, provably eternal looping, and $N(n - N)$ free parameters: at $\rho = 0.95$ about 5×10^4 , at $\rho = 0.5$ about 2.5×10^5 . The rank fraction ρ is thus the single dial trading information density against aesthetic freedom — the playlist artwork of the earlier sections is the extreme low- ρ case, the pixel-exact song print the extreme high- ρ case.

The artwork in one line

$$\mathbf{A} = \underbrace{\mathbf{X} \mathbf{S} \mathbf{G}^{-1} \mathbf{X}^\top}_{\text{plays the playlist}} + \underbrace{\mathbf{U}(\mathbf{P}_\perp \mathbf{V})^\top}_{\text{draws the picture}}, \quad \mathbf{A}\mathbf{x}_n = \mathbf{x}_{n+1} \text{ exactly}, \quad \mathbf{\Pi} \mathbf{A} \mathbf{\Pi}^\top \approx \mathbf{T}.$$